Circle K Homework Case

Abrahim Borgi

Contents

Case 1 2
Introduction 2
Key Findings 3
Overall conclusion 4
Approach 5
Data Model & Preparation 5
Data Model 6
Measures 7
Report Pages 7
Common Slicers 7
Explanation Boxes 8
Page 1: Overview 8
Page 2: Compare Sites 9
Page 3: Driver Tree 11
Case 2 12
Introduction 12
Conclusion 13
Overall conclusion 13
Analytical Approach 14
Limitations and “Potential” Error 15
Additional Material 15

Case 1

Introduction

The purpose of this case study was to analyze ice-cream sales across Danish sites and evaluate performance between 2017 and 2018, with a focus on the impact of a 2018 promotion.

I designed three report pages with different purposes:

1. **Overview** – high-level KPIs (sales, margin, transactions, top products, SPLY).
2. **Compare Sites** – A/B testing between stations with statistical significance.
3. **Driver Tree** – decomposition of Sales Margin to show key drivers of change.

As part of my application, I had already built a full **Circle K simulation project**, so here I focused on adding **new elements** (especially in page 2 and 3) to show a broader range of skills.

Key Findings

Overall

- In 2018, both total sales and sales margin increased compared to 2017. Sales grew by around **28%**, while sales margin grew by about **16%**.
- The main driver of this improvement was **higher total sales**, which came from growth of the **average gross amount**, which came from higher **number of units** in avg. being sold.
- The increase was not evenly spread. **Station 3** and **4** showed strong growth, while other sites contributed less consistently. **Station 5** was still the best-selling site but without a large growth.

Ice-Cream Promotion

- The 2018 ice-cream promotion created a **clear uplift in both sales and sales margin**. With promo included, sales margin reached **90,009 kr. (+16% vs. SPLY)** compared to **83,372 kr. (+7% vs. SPLY)** without promo.
- The uplift is mainly explained by **basket size**:
    - **Average number of units per transaction** rose sharply (**+33% with promo** vs. only **–1% without promo**).
    - **Average price per unit** dropped (**–5% with promo** vs. **+18% without promo**), showing customers bought more but cheaper units.
    - Combined, this pushed **average gross amount up by +27% with promo**, compared to just **+17% without promo**.
- On the cost side, the promotion also led to significantly higher expenses: **Total cost price rose +36% with promo** vs. only **+6% without promo**, which limited the margin growth relative to the total sales increase.
- The strongest promo effect was seen at **Station 1 and Station 3**.
- The promo's biggest impact period was **24 July – 4 August**.

KPI	With Promo (SPLY)	Without Promo (SPLY)
Sales Margin	**+16%** (90,009 kr.)	**+7%** (83,372 kr.)
Total Sales	**+28%** (331,799 kr.)	**+6%** (274,789 kr.)
Total Expenses	**+34%** (241,790 kr.)	**+6%** (191,417 kr.)
Average Gross Amount	**+27%** (72.38 kr.)	**+17%** (66.39 kr.)
Number of Transactions	**+1%** (4,584)	**–9%** (4,139)
Avg. Units/Transaction	**+33%** (3.76)	**–1%** (2.79)
Avg. Price per Unit	**–5%** (19.23 kr.)	**+18%** (23.79 kr.)
Total Cost Price	**+36%** (175,430 kr.)	**+6%** (136,459 kr.)

Overall conclusion

In summer 2018, sales grew **28%** and sales margin **16%** vs. 2017, mainly from **larger baskets (more units per transaction)**.

The ice-cream promotion played a key role in this uplift. It peaked between **24 July – 4 August**, where sales spiked well above the prior year, and had the strongest effect at **Station 1** and **Station 3**.

The promo boosted volumes and gross amounts significantly, though it also led to **higher expenses and cost prices**, which tempered the margin gains compared to the sales increase.

Approach

To solve the case, I combined a structured **data foundation** with clear **analytical pages** in Power BI.

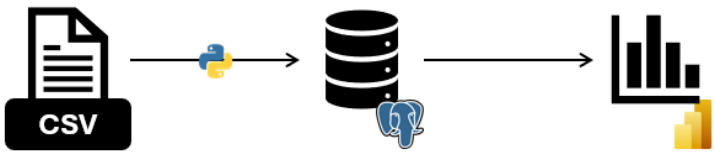
The process can be summarized in two parts:

- **Data preparation & modeling**
  - Loaded raw CSV files into **PostgreSQL**, applied SQL views for structuring.
  - Imported curated tables into **Power BI**, building a clean **star schema**.
  - Created **measures**.
- **Analysis & visualization**
  - Designed three report pages, each with a distinct purpose:
    - **Overview** – high-level KPIs, trends, and top products.
    - **Compare Sites** – A/B comparisons of stations.
    - **Driver Tree** – breakdown of selected KPI into its main drivers.
  - Common **slicers** across pages (metric, promo, region, date) ensured interactive and consistent exploration.

This approach allowed me to keep heavy transformations in SQL while showcasing business insight and visualization skills in Power BI.

Data Model & Preparation

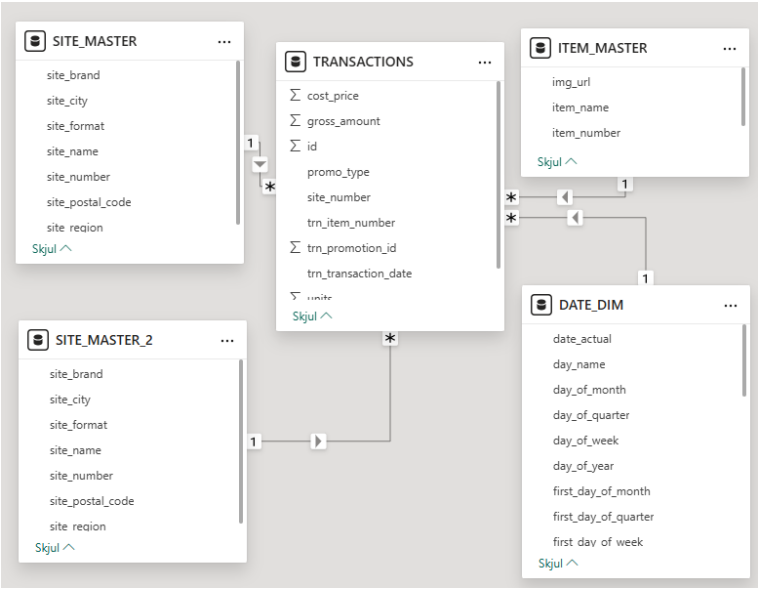
To prepare the analysis, I first ingested the CSV files into PostgreSQL, applied some SQL views for structuring, and then loaded the curated tables into Power BI.



This way, **the heavy lifting stayed in SQL** while Power BI was kept clean for business logic and visuals.

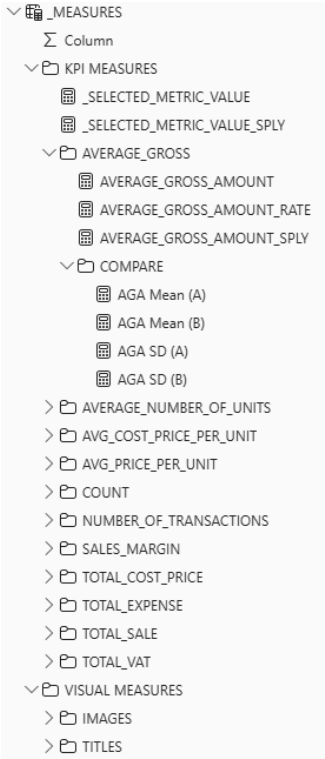
Data Model

- Central **TRANSACTIONS** fact table with gross amount, cost, VAT, units, and promotion flag.
- Connected to:
  - **DATE_DIM**: Enables year-over-year and SPLY comparisons + filtering.
  - **ITEM_MASTER**: Product names, categories, and images.
  - **SITE_MASTER**: Site-level details such as region, city, and brand.
- Modeled as a **star schema** for simplicity and performance.



Measures

In Power BI, I grouped all measures into clear folders to support different parts of the report:



- **Folder: KPI Measures**: core metrics like *Sales Margin, Total Sale, Total Expense, Total VAT, Total Cost Price, Number of Transactions, Average Gross Amount, Average Price per Unit, Average Cost per Unit*.

  Also includes the **_SELECTED_METRIC_VALUE** and **_SELECTED_METRIC_VALUE_SPLY** used for the dynamic metric selector.

- **KPI Measures \ Compare:** Measures for mean and standard deviation across Site A / Site B groups. These are used specifically on the *Compare Sites* page.

- **Folder: Visual Measures** – titles and image URLs used for dynamic labeling and visuals.

This structure keeps Overview and Driver Tree powered by standard KPI measures, while Compare Sites has its own dedicated set under **Compare**, ensuring the logic stays tidy and reusable.

Report Pages

Common Slicers

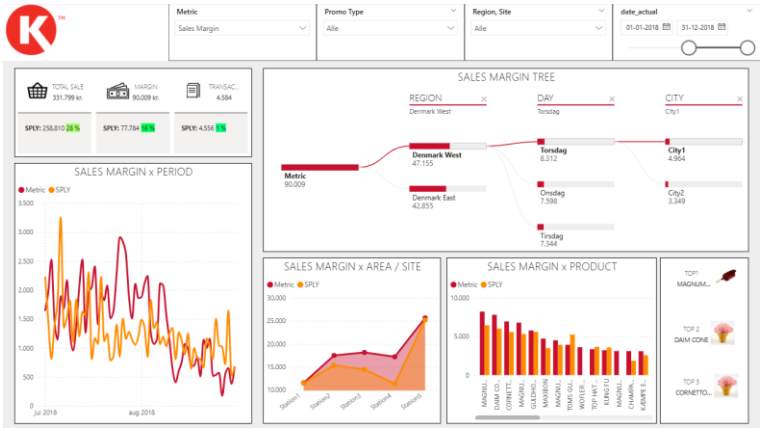
Across all pages I included consistent slicers so the user can easily filter the analysis:

- **Metric selector** – drives which KPI is shown in the visuals.
- **Promo type** – filter on transactions with or without promotions.
- **Region / Site** – drill into specific areas or individual stations.
- **Date range** – timeline filter to select relevant periods.

Explanation Boxes

To make the report more self-explanatory, each page includes an **HTML-based explanation box**. These boxes can be expanded or collapsed as dropdowns.

Page 1: Overview



Purpose: Provide a high-level introduction to the dataset with the main KPIs, trends over time, and top products.

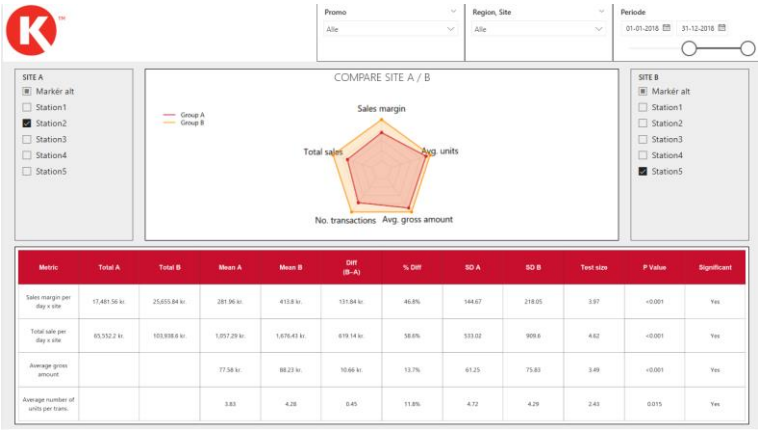
Visualisations:

- KPI cards for total sales, margin, and transactions.
- Charts for KPI vs. SPLY overtime for selected metric.
- Breakdown chart (Tree chart) that allows the user to flexibly drill down the selected metric by chosen dimensions (e.g., region, city, item) instead of being locked to a fixed hierarchy.
- Top 3 selling product with ”images” (Not real product images).

Key questions this page can answer

- How have total sales and sales margin developed compared to last year?
- Which products are the top sellers?
- How do sales break down by underlaying dimension?

Page 2: Compare Sites



Purpose: Allow direct comparison between two sets of sites.

Special selectors:

- **Site A selector** – choose one or multiple sites for group A.
- **Site B selector** – choose one or multiple sites for group B.

Visualisations:

- Radar chart comparing KPI profiles between Group A and Group B.
- Table with totals, averages, differences, % differences, standard deviations, test sizes, p-values, and significance indicators for four key metrics:
    1. Sales margin per day × site
    2. Total sale per day × site
    3. Average gross amount
    4. Average number of units per transaction

Key questions this page can answer

- Are differences in sales margin or total sales between two stations statistically significant?
- Do transaction sizes or unit counts differ between sites?

- How large is the gap in performance, and is it consistent across metrics?

Note:

To run the visualisations in Power BI, you need to have **R installed** on your machine and connected to Power BI.

You also need the following R packages:

- **dplyr**
- **tidyr**
- **fmsb**
- **gridExtra**

You can find an R script that installs these packages in the *Circle-K Interview* project under *source\workspace\requirement_R_packages.R* next to the PBIP file.

Page 3: Driver Tree

Purpose: Show the breakdown of Sales Margin into its underlying drivers in a clear, top-down view.

Visualisations:

- **Driver tree** (all KPI cards combined into one collective visualization) that dynamically adapts to the chosen metric.
  - Each KPI can be decomposed into its main drivers (e.g., Sales Margin → Sales + Expenses; Sales → Transactions × Avg. Gross Amount; Avg. Gross Amount → Units × Price).
  - The tree highlights both absolute values and YoY changes (SP/SPLY), making it easy to see what drive improvements or declines.
- Supporting trend and area charts.

Key questions this page can answer

- Which factors are driving changes in Sales Margin?
- Is growth driven more by higher transactions or by pricing effects?

Case 2

Introduction

Question: How did this communication perform?

To evaluate the campaign, we set up an **A/B test** where one group of members (target) received communication, while a smaller control group did not. By comparing the two groups, we can isolate the effect of the campaign.

This leads to three underlying questions that guide how we measure performance in practice:

1. **Did the campaign increase the share of members who fueled?**

Measured as the **conversion rate** i.e. proportion of customers in each group who fueled at least once.

2. **Did the campaign change the average fueling amount among those who fueled?**

Measured as the **average liters per fueler**.

3. **Did the campaign increase overall fuel volume per member?**

Measured as the **average liters per member.**

Together, these three KPIs provide a structured way to assess whether the campaign led to more customers fueling and if these customer fueled more.

Conclusion

The campaign was evaluated through an A/B test with three KPIs:

- **KPI 1: Did more members fuel?**
  - Conversion rate rose from **12.8% → 14.2%** (+1.4 pp, +11%).
  - Statistically significant (**p = 0.0052**).
- **KPI 2: Did fuelers buy more liters?**
  - Avg. liters per fueler increased from **27.1 L → 30.4 L** (+3.3 L, +12%).
  - Statistically significant under reasonable variance assumptions (**p = 0.0030**).
- **KPI 3: Did overall fuel per member increase?**
  - Avg. liters per member rose from **3.48 L → 4.33 L** (+0.85 L, +24%).
  - Highly statistically significant (**p ≈ 0**).

KPI	Control	Target	Target vs. Control	Significant?
KPI 1 - Conversion rate	12.8%	14.2%	+1.4 pp (+11%)	Yes (p = 0.0052)
KPI 2 - Avg. liters per fueler	27.1 L	30.4 L	+3.3 L (+12%)	Yes (p = 0.0030)
KPI 3 - Avg. liters per member	3.48 L	4.33 L	+0.85 L (+24%)	Yes (p ≈ 0)

Overall conclusion

All three KPIs show a clear and statistically significant uplift. The campaign increased both the share of members who fueled, and the liters purchased per fueler/member.

In short, the campaign **successfully drove incremental fuel sales**.

Analytical Approach

To evaluate the campaign, we set up an **A/B test** with a target group (received communication) and a control group (no communication).

The three KPIs were tested as follows:

- **KPI 1: Conversion rate**
  - Outcome: whether a member fueled at least once (yes/no).
  - Tested using a **Binomial model** (two-proportion test).
  - Appropriate because each member either fuels or does not.
- **KPI 2: Average liters per fueler**
  - Outcome: liters purchased among members who fueled.
  - Here we only had aggregated totals, not individual data.
  - We therefore assumed that liters per fueler follow a **Normal distribution** and set a reasonable "coefficient of variation" (CV) to describe variability.
  - This allowed us to estimate significance, and we checked robustness by testing different CV levels (e.g. 0.8, 1.0, 1.2), all with same result.
- **KPI 3: Average liters per member (incl. non-fuelers)**
  - Outcome: liters per member across the whole population (fuelers + non-fuelers).
  - Technically, we already expected this result to be significant since both conversion (KPI 1) and liters (KPI 2) grew significantly.
    If both components increase, the combined measure inevitably will.
  - Still, we modeled it explicitly as a **mixture**: a Bernoulli distribution (0/1 outcome) and a Normal distribution for liters.

- This confirms the effect was highly significant and robust to different variance assumptions.

All analyses and calculations were performed in **R**, where the testing framework and assumptions are documented.

Limitations and "Potential" Error

A few limitations of this analysis are worth mentioning:

- **Aggregate data for KPI 2 and 3**
  - We only had totals, not individual fueling data.
  - This required assuming a Normal distribution and an estimated variance (CV)
- **Missing data and subgroup effects**
  - Without detailed breakdowns (e.g., by station, geography, time of day), we cannot rule out hidden patterns.
  - For example, a **Simpson's paradox** effect could appear if the campaign worked very well in some segments but not at all in others.

Overall, I do not think these limitations change the conclusions cause the results are so clear.

Additional Material

- The **entire project** with all relevant code can be found here:
  https://github.com/AbrahimBorgiPrivat/Circle-K---Interview---project
  (the repository will remain public until Wednesday)
- The **Power BI file** will be sent directly, and can also be found in the GitHub repo under:
  source > workspace
- The **R code for Case 2** is available in the GitHub repo under:
  source > code > case2 > R